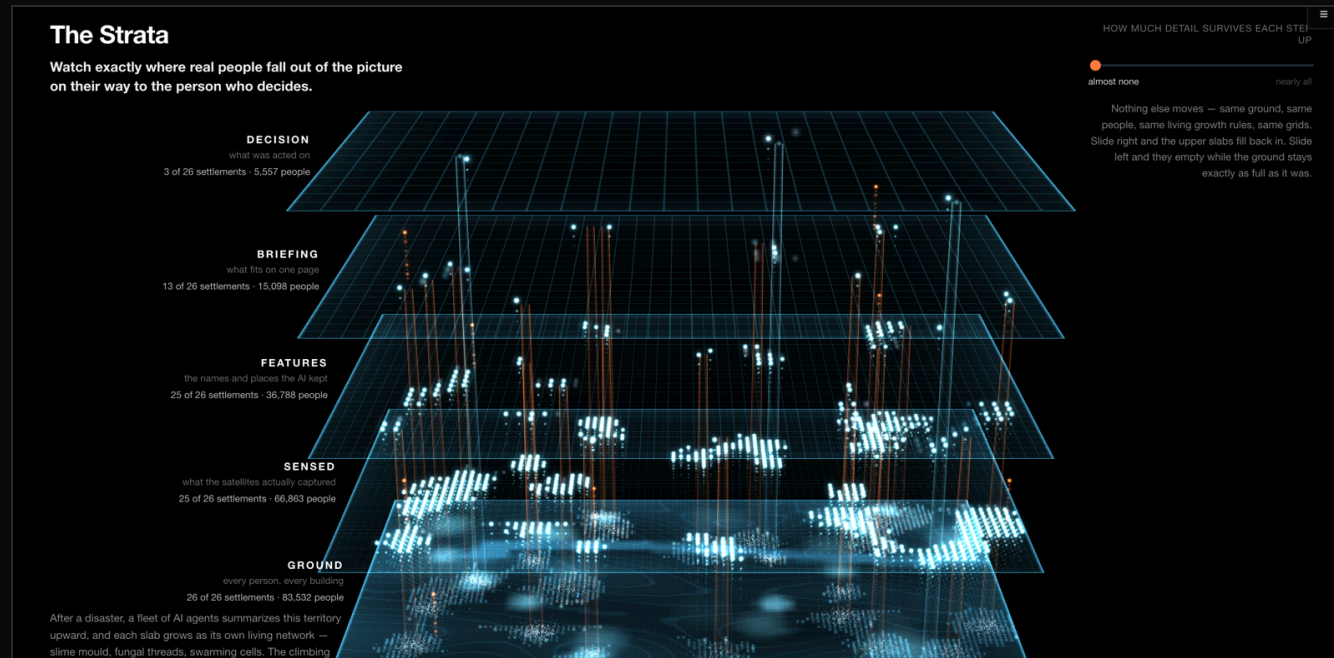


We build simulated disasters and hand the rescue work to fleets of AI agents. Each fleet is organized the way a living thing grows, and that growth rule decided who got seen, proven safe, and saved.



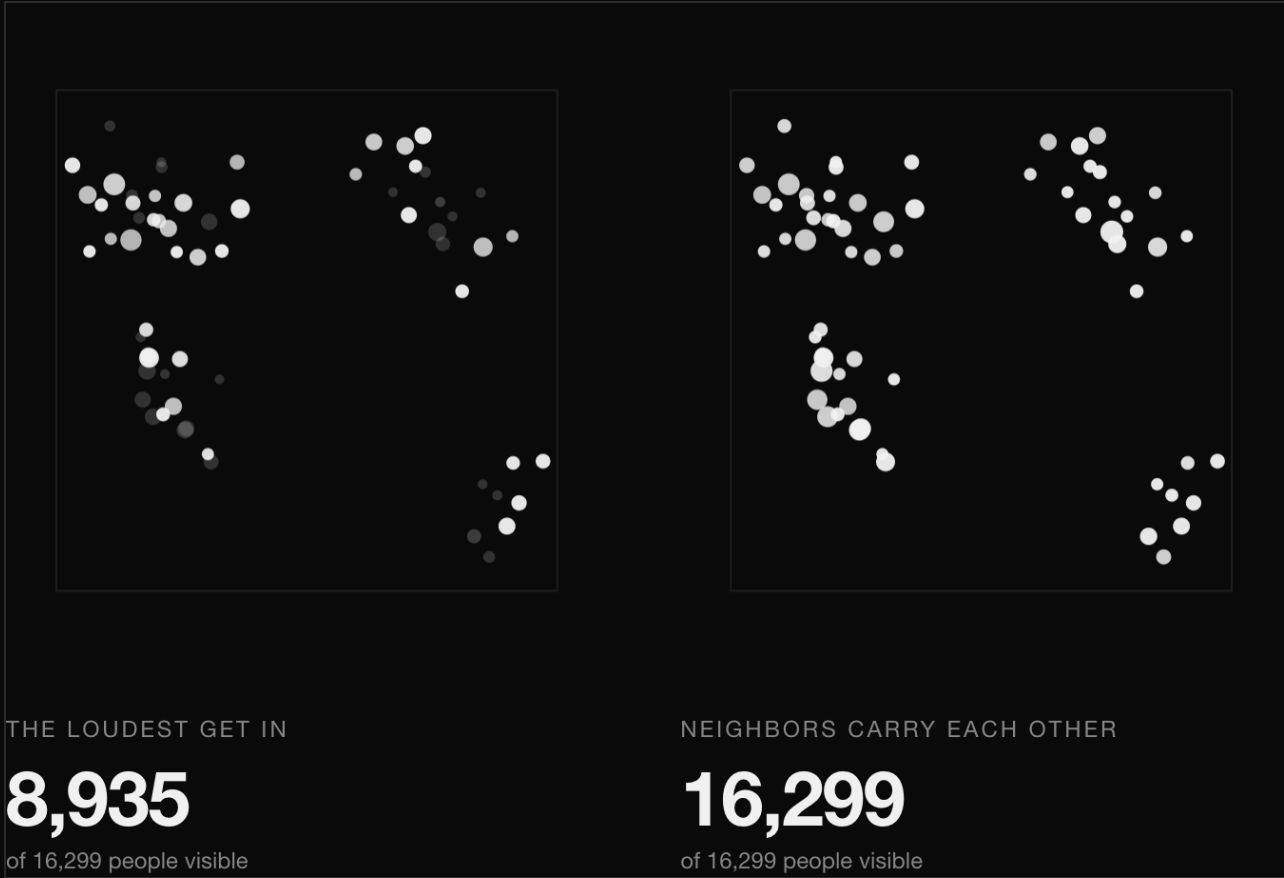
Watch exactly where real people fall out of the picture on their way to the person who decides.

The same flooded territory, five summarizing layers high. The bottom layer holds every person and building. The top layer holds what the decision office acts on. A settlement's column of light stops where the fleet lost track of it.

Predictions are saved publicly before anything runs, and scripts count every score.

- Every world is generated from a committed seed, and the same seed produces byte-identical event logs, asserted on every run.
- Exactly one rule differs between compared runs — the growth rule under test — and budgets are matched and asserted in code.
- The organizing rules are enforced mechanically by the harness. The AI agents only read, judge, and choose.
- Scoring is a script run against an answer key sealed before the experiment. Nobody grades by opinion.
- Thirteen of our own pre-registered predictions have been refuted so far, and every refutation is published beside the survivors.

A flood hit 64 communities. The way reports were summarized decided who got help.



26 of 64

communities erased from the briefing under the standard crowding rule, while survey visits stayed perfect

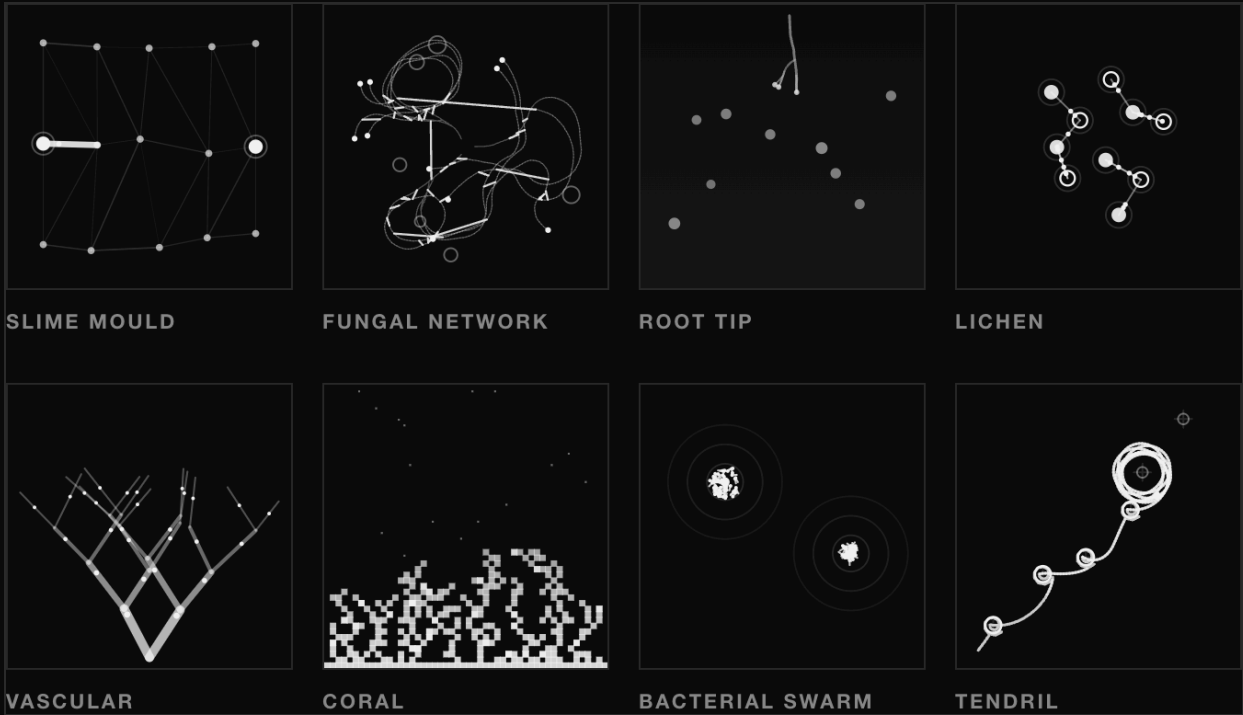
64 of 64

communities kept visible by the fungal-network merging rule, at the same cost

8,935 → 16,299

people the decision office could see, before and after the rule change

Eight growth rules from living networks, translated into exact, runnable code.



- SLIME MOULD**

feed the busy paths, abandon the quiet ones
- FUNGAL NETWORK**

connect neighbors so they share and survive
- ROOT TIP**

go get the missing fact directly
- LICHEN**

pair two partners with opposite weaknesses
- VASCULAR**

one trunk, ranked branches, locked tips
- CORAL**

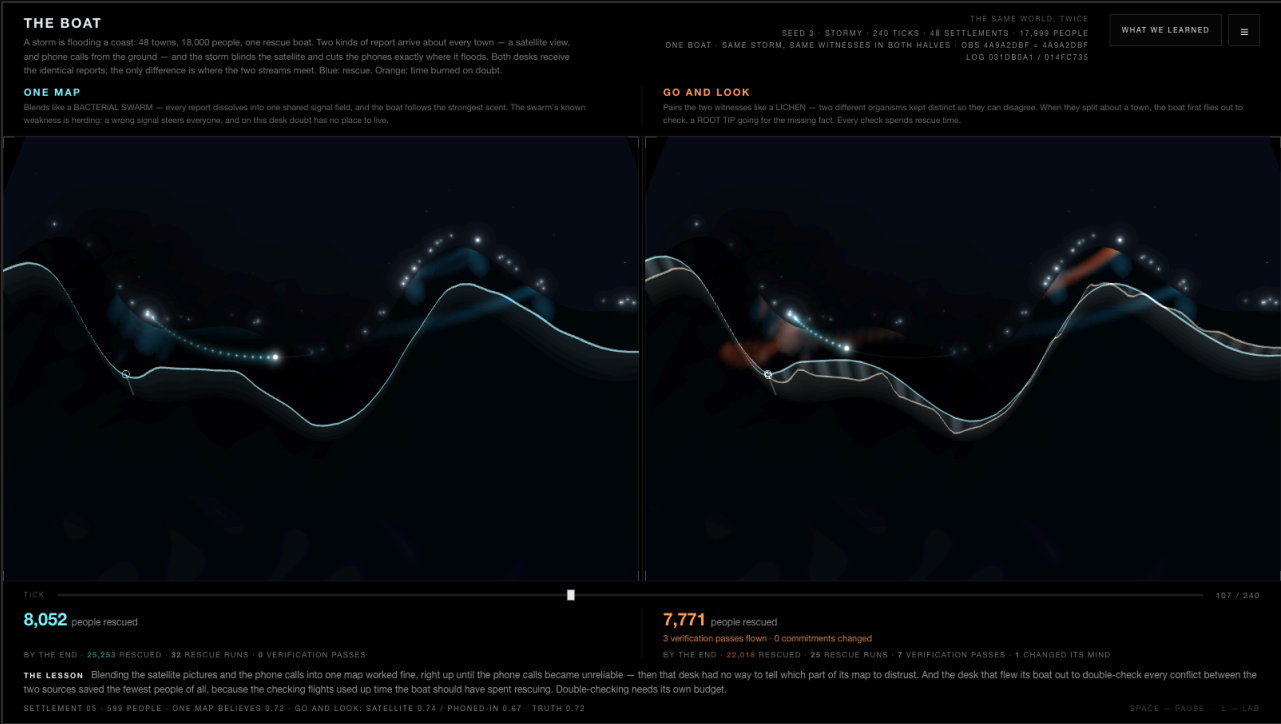
add layer on layer, erase nothing
- BACTERIAL SWARM**

herd toward the strongest signal
- TENDRIL**

sweep until you find support, then wrap it

Seven of the eight have been through real experiments. The organizing rule runs as harness machinery; the agents do the thinking.

Phone callers lied about a flood, and the fleet forced to double-check reached a third fewer people.



Half a bridge-inspection office was destroyed, and only the team that kept records of its discards stayed perfect.

THE BRIDGE

A city's only bridge is wired with 256 sensors, and one office decides every morning whether it is safe to cross. No person can read 256 raw readings, so computers summarize them in stages: 256 readings become 64 reports, then 16 findings, then 8 claims — one per section of the bridge — then the one sentence on the desk. THE BRIDGE IS SAFE. 21,000 people cross on that sentence. You are watching the summarizing system built twice on the same budget, under two different rules for which supporting links get kept. The same 10 computers fail in both towers, and we measure one thing: for each claim still on the desk, can an engineer still trace it down to a real sensor reading? Blue: a claim an engineer can still trace. Orange: a claim still being used, with nothing checkable underneath it.

THE STRONGEST SUPPORT WINS

Grows like BLUE MOULD: find the busiest path, let every other link die, keep no record of what was dropped. One thick spine — and when it breaks, nothing remembers how to rebuild it.

RECORDS KEPT AT PRUNING: 0 — NOTHING TO REPLAY

THE SAME CONCLUSION, BUILT TWICE

SEED 5 · 345 NODES · 21,000 PEOPLE · 156 OF 166 EDGE-UNITS SPENT IN EACH STACK

THE SAME 10 COMPUTERS FAIL IN BOTH · LOGS 8A689474 / 4CC7500D

TOP TO BOTTOM: THE SENTENCE · CLAIMS · FINDINGS · READINGS · THE GROUND

WHAT WE LEARNED

NEIGHBORS CARRY THE PROOF

Grows like a FUNGAL NETWORK: the same budget spreads across several neighbors, and — the CORAL rule — every cut link leaves a tombstone, a tiny record. When damage comes, a repair pass replays the records.

RECORDS KEPT AT PRUNING: 524 TOMBSTONES · 67 CROSS-LINKS

10 of the 32 computers doing the summarizing have just failed — the same 10 in both towers.

ACT 2 OF 5

156 of 156 edge-units wired · 23 of 32 computers alive

BREAKS 4 · TOMBSTONES 0 · HEALS 0 — PERMANENT

THE LESSON We damaged both towers of summaries the same way. The tower that had kept a short note about every sensor connection it ever discarded used those notes to rebuild every broken path from its conclusions down to real readings. The tower that kept no notes rebuilt nothing — 14,420 people were left crossing the bridge on conclusions that can no longer be checked against any sensor. When a system throws information away, make it write down what it threw.

CLAIM 343 · 3,491 PEOPLE · THE WALKS BEGIN AT T140

156 of 156 edge-units wired · 23 of 32 computers alive

BREAKS 2 · TOMBSTONES 524 · HEALS 0 OF 2

SPACE — PAUSE · L — LAB · CLICK A CLAIM TO TRACE IT

128 / 200

8 of 8

safety statements still provable after damage for the team that recorded every discarded source

7 of 8

for the single-source team — the predicted collapse failed because real summarizers carry sources in their prose

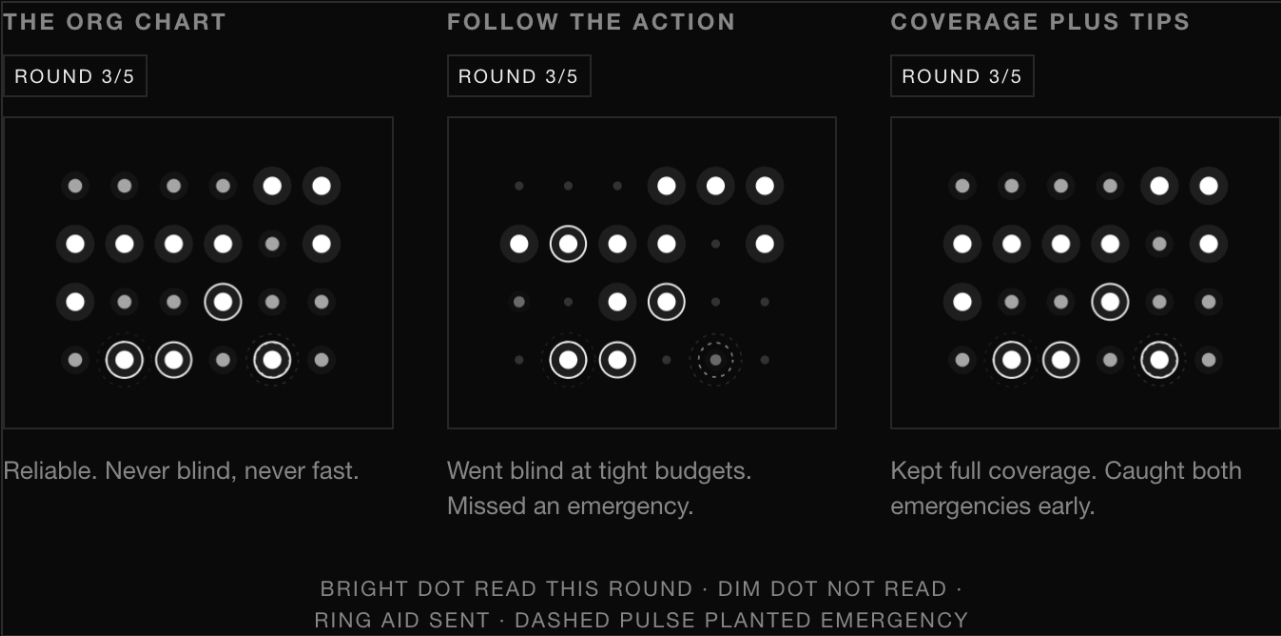
524

tombstones — small records of every cut — held by the record-keeping tower when the damage came

AURAWORLD

RECORDS: DOCS/TRUST/THE-BRIDGE/LIVE · PAGE 6 OF 8

With slack, every organization saved everyone. At a tight budget, the organizing rule decided which emergencies got caught.



5 of 5, three ways

deadline emergencies served by every rerouting rule in the deadline region — and one dispatcher legally shortened its routes to dodge its own commitment rule

12 of 24

towns the slime-mould fleet could still see at a 50 percent reading budget; it left an erupted emergency unread for three rounds

20 of 24

towns held by the fungal fleet, whose chief aided both emergencies before they erupted

Growth rules decide the outcome when the budget binds. We want the compute to find exactly where each rule stops being safe.

What we found at small scale. Where a rule truly forced the agents' hands, the simulated failures replicated exactly with real agents. Where any slack remained, capable agents rescued every structure. The choice of organizing rule is cheap insurance that pays out precisely when capacity is scarce.

What scale unlocks. Thresholds — the exact budget where each rule flips from asset to liability — and the monoculture question: do seven institutions sharing one model fail together, where seven genuinely different model families catch each other?

~1,000,000

agent decisions planned: fleets of 100 to 300 agents, 5 to 10 rounds, 3 to 5 rules, 10 seeded trials, about 25 budget configurations

7 × 24 GB

one local model server per card (Qwen3 32B and 8B, Gemma 3 12B, Mistral Small 24B, served by vLLM); full 168 GB needed for the seven-family monoculture run

40–60 h

expected usage across the competition window, from smoke runs to threshold sweeps to repeats